

SELENE: Selective-Evaluation Lattice Encryption with Norm Equilibrium

A Single Unified Post-Quantum Scheme Beyond the GPV-CKKS Impossibility

Module Inner-Product Functional Encryption from MLWE -- Trapdoor-Free, Small-Key, IND-Secure

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Abstract

This paper presents SELENE, the first post-quantum encryption scheme explicitly constructed as a *direct constructive response* to the GPV-CKKS impossibility theorem [QuantumVault Thesis, Theorem 8.1]. That theorem proves that GPV-style adaptive HIBE and CKKS-style approximate FHE cannot coexist in a shared ring R_q because the key norm lower bound $LB = n \cdot \sqrt{k} \cdot \sqrt{q} \cdot \omega(2 \cdot \sqrt{2})$ from SampleLeft incompatibly exceeds the CKKS upper bound $UB = \sqrt{n}$. SELENE sidesteps this impossibility by abandoning both GPV trapdoors and CKKS arithmetic entirely, achieving the same *functional goals* -- identity-based access and computation over encrypted data -- through a single new algebraic mechanism: Module-LWE Inner-Product Functional Encryption (MLWE-IPFE). SELENE provides: (1) identity-based decryption as a special case of inner-product evaluation; (2) secret keys with norm $\|sk_f\|_2 = O(\|f\|_1 \cdot \sqrt{k \cdot n \cdot \eta})$ -- drawn from the standard CKKS secret distribution χ_s , satisfying Theorem 7.1 of the companion thesis; (3) no SampleLeft trapdoor (avoiding the LB constraint); and (4) IND-FE security with a tight reduction to $MLWE_{\{n,k,q,\chi\}}$. We formally verify that SELENE satisfies none of M1, M2, or M3 of Definition 5.1, placing it entirely outside the impossibility scope while delivering equivalent practical utility. Three security theorems are proved. A Python reference implementation validates correctness. Parameters are given for NIST Levels 1, 3, and 5.

Keywords: Module-LWE, Functional Encryption, Inner-Product, Post-Quantum, IND-Security, No-Trapdoor, Norm-Compatible, GPV-CKKS Impossibility, QuantumVault

1. Introduction and Motivation

The companion thesis [QV-Thesis] proves a fundamental impossibility: no scheme satisfying all three of Definition 5.1 -- (M1) shared ring $R_{\{q_FHE\}}$, (M2) GPV-adaptive HIBE via SampleLeft, and (M3) CKKS-style approximate FHE -- can exist for any ring dimension $n \geq 3$ under standard MLWE and RLWE hardness. The impossibility arises from a direct key-norm collision:

The Impossibility Gap (from QV-Thesis, Theorem 8.1).

Lower bound from SampleLeft (Lemma 6.1): $\|sk\|_2 \geq LB = n \cdot \sqrt{k} \cdot \sqrt{q} \cdot \omega / (2 \cdot \sqrt{2})$

Upper bound from CKKS-RLWE reduction (Theorem 7.1): $\|sk\|_2 \leq UB = \sqrt{q} \cdot \sqrt{n}$

Feasibility condition $LB \leq UB$ requires $n \cdot k \cdot q \cdot \omega^2 \leq 8$. Under MLWE hardness ($q \geq n$), $LHS \geq n^2 \cdot \log_2(n) \gg 8$ for all $n \geq 3$.

At Kyber-768 ($n=256, k=3, q_{FHE}=2^{40}$): $LB/UB > 29,000,000$. Post-quantum parameters strictly WIDEN this gap.

This impossibility is precise and tight. The natural question it raises is not 'can we circumvent it?' -- the companion thesis proves we cannot for $M1+M2+M3$ -- but rather: *can we achieve the same practical goals through different mathematics?* The goals are (i) identity-based access control over encrypted data, and (ii) computation over ciphertexts without decryption. SELENE answers this question affirmatively, achieving both goals through a single new construction that violates all three conditions $M1, M2, M3$ simultaneously and provably so.

1.1 The SELENE Idea in One Paragraph

Core Mechanism: Inner-Product Functional Encryption over Module Lattices.

SELENE replaces the GPV trapdoor ($M2$) with inner-product key derivation: for N encrypted slots $x_1, \dots, x_N$, a key for function vector $f = (f_1, \dots, f_N)$ in $\mathbb{Z}^N$ is simply $sk_f = \sum_{i=1}^N f_i \cdot s_i$ -- a SHORT linear combination of N independent MLWE secrets. This key allows computing the inner product over encrypted data. Identity-based access is the special case $f = e_{id}$ (unit vector): $sk_{\{e_{id}\}} = s_{id}$ decrypts slot id exclusively. No SampleLeft, no gadget trapdoor, no large keys. Key norm: $\|sk_f\|_2 \leq \|f\|_1 \cdot \max_i \|s_i\|_2 = O(\|f\|_1 \cdot \sqrt{k \cdot n \cdot \eta})$ -- directly from the RLWE/MLWE secret distribution χ_s (satisfying Theorem 7.1 of QV-Thesis). CKKS arithmetic ($M3$) is unnecessary: SELENE evaluates inner products exactly via the decryption formula, not approximately via FHE circuit evaluation.

1.2 Contributions

C1 -- SELENE Construction (Section 4): A complete Module-LWE Inner-Product Functional Encryption scheme: Setup, KeyGen, Enc, Dec. No trapdoors. Keys from standard MLWE distribution χ^k .

C2 -- Theorem 1 (IND-FE Security, Section 5.1): Tight IND-FE reduction to $MLWE_{\{n,k,q,\chi\}}$ via N -game hybrid argument. Proof is self-contained, references no prior FE results.

C3 -- Theorem 2 (Key Norm Bound, Section 5.2): Explicit norm bound $\|sk_f\|_2 \leq \|f\|_1 \cdot \sqrt{k \cdot n \cdot \eta}$. For bounded functions $\|f\|_1 \leq B$: $\|sk_f\|_2 \leq B \cdot \sqrt{k \cdot n \cdot \eta} = O(\sqrt{n})$. This directly satisfies Theorem 7.1 of QV-Thesis (key in $\text{supp}(\chi_s)$).

C4 -- Theorem 3 (Impossibility Exemption, Section 5.3): Formal proof that SELENE satisfies none of $M1, M2, M3$ of Definition 5.1 [QV-Thesis]. SELENE is provably outside Theorem 8.1 scope.

C5 -- IND-CCA2 Extension (Section 5.4): FO^{perp} -style CCA2 upgrade for single-key decryption. Reduces to IND-FE + MLWE in ROM.

C6 -- Reference Implementation (Section 7): Python implementation with correctness self-test. All algorithms match paper specification.

1.3 What SELENE Is and Is Not

SELENE provides:

(a) Post-quantum IND-FE security from $\text{MLWE}_{\{n,k,q,\chi\}}$. (b) Identity-based exclusive decryption (unit-vector keys). (c) Inner-product evaluation over N encrypted slots. (d) Keys satisfying the CKKS norm bound (Theorem 7.1, QV-Thesis). (e) No trapdoors, no large Gaussian parameters, no `SampleLeft`.

SELENE does NOT provide:

(a) Full FHE (only bounded linear functions, not arbitrary circuits). (b) GPV-adaptive hierarchical key delegation (no `SampleLeft` = no hierarchy). (c) Approximate arithmetic (decryption is exact, not CKKS-style). These limitations are honest: SELENE achieves a STRICT SUBSET of the impossible combination's goals, but through a provably secure mechanism.

Novelty claim (precisely scoped):

To the best of our knowledge, SELENE is the first scheme explicitly designed as a constructive counterpart to the GPV-CKKS impossibility, with a formal proof that its keys satisfy the CKKS norm bound (Theorem 2) and a formal proof that it is outside Definition 5.1 (Theorem 3). The MLWE-IPFE construction adapts LWE-based IPFE [ALS16, BJKL18] to the module setting with tighter proofs and explicit norm analysis.

2. The GPV-CKKS Impossibility -- Precise Scope

We restate the impossibility from QV-Thesis precisely, to enable the formal exemption proof in Section 5.3. All definitions and theorems here are from the companion thesis.

Definition 2.1 (FHE-Native HIBE, QV-Thesis Def. 5.1).

A scheme FN-HIBE satisfies Definition 5.1 iff it satisfies ALL of:

M1 (Shared Ring): The HIBE and FHE components share ring $R = R_{\{q_FHE\}}$ with the same modulus q_FHE .

M2 (GPV-Adaptive HIBE): IND-ID-CPA security in the standard model via the `SampleLeft` simulation of [CHKP10]. Adaptive key queries are answered by `SampleLeft` with Gaussian parameter $\sigma \geq \sqrt{n \cdot q_FHE / 2} \cdot \omega$ (Lemma 6.1, QV-Thesis).

M3 (CKKS Approximate FHE): IND-CPA security provable by reduction to $\text{RLWE}_{\{n,q,\chi_s,\chi_e\}}$ with χ_s the standard CKKS secret distribution. This forces $\|sk\|_2 \leq B_s = O(\sqrt{n})$ (Theorem 7.1, QV-Thesis).

Theorem 2.1 (GPV-CKKS Impossibility, QV-Thesis Theorem 8.1).

No scheme satisfying Definition 2.1 (M1 + M2 + M3) exists for $n \geq 3$ under Assumptions A1 (MLWE hardness), A2 (RLWE hardness), A3 ($q_{\text{FHE}} \geq n$).

The impossibility arises from: $LB = n \cdot \sqrt{k} \cdot \sqrt{q_{\text{FHE}}} \cdot \omega / (2 \cdot \sqrt{2}) > \sqrt{n} = UB$

since $n \cdot k \cdot q_{\text{FHE}} \cdot \omega^2 / 8 \geq n^2 \cdot \log_2(n) / 8 > 1$ for all $n \geq 3$.

The impossibility is specific and structural: it applies to ALL schemes satisfying M1+M2+M3. Schemes that violate ANY of M1, M2, M3 are not covered. SELENE is designed to violate all three simultaneously, while achieving the PRACTICAL GOALS motivating the impossible combination.

3. Mathematical Preliminaries

3.1 Module-LWE

Definition 3.1 (MLWE).

Let $R_q = \mathbb{Z}_q[x]/(x^{n+1})$, n a power of 2, q prime, $k \geq 1$, χ the error distribution over R_q . $\text{MLWE}_{\{n,k,q,\chi\}}$: given (A, b) with $A \leftarrow U(R_q^{k \times k})$ and $b = A*s + e$ for $s, e \leftarrow \chi^k$, distinguish from (A, u) with $u \leftarrow U(R_q^k)$.

$$\text{Advantage: } \text{Adv}_{\text{MLWE}}(B) = |\Pr[B(A, A*s+e) = 1] - \Pr[B(A, u) = 1]|$$

Definition 3.2 (B_{eta}). $B_{\text{eta}} = \sum_{i=1}^k \text{eta} (a_i - b_i)$ for $a_i, b_i \leftarrow \text{Bernoulli}(1/2)$. Mean 0, variance $\text{eta}/2$, support $[-\text{eta}, \text{eta}]$. Each coefficient c of $s \leftarrow B_{\text{eta}}^k$ satisfies $E[c^2] = \text{eta}/2$; $\|s\|_2^2 = \sum$ of $n*k$ coefficients, so $E[\|s\|_2^2] = n*k*\text{eta}/2$.

Lemma 3.1 (Key Norm, $\chi = B_{\text{eta}}$). For $s \leftarrow B_{\text{eta}}^k$ in R_q^k : $\|s\|_2 \leq \sqrt{n*k} * \text{eta}$ (worst-case, all coeffs = $\pm \text{eta}$). Typical (expected) norm: $\|s\|_2 \sim \sqrt{n*k*\text{eta}/2}$. With probability $\geq 1 - 2^{-n}$: $\|s\|_2 \leq \sqrt{n*k*\text{eta}}$. [Markov + sub-Gaussianity]

3.2 Inner-Product Functional Encryption

Definition 3.3 (IPFE Syntax).

An Inner-Product Functional Encryption scheme $\text{IPFE} = (\text{Setup}, \text{KeyGen}, \text{Enc}, \text{Dec})$ for message space $M = R_q^N$ and function space $F = \mathbb{Z}^N$ satisfies:

$\text{Setup}(1^\lambda, N) \rightarrow (\text{mpk}, \text{msk})$: generates master key pair for N slots.

$\text{KeyGen}(\text{msk}, f) \rightarrow \text{sk}_f$: for f in F , generates evaluation key.

$\text{Enc}(\text{mpk}, x) \rightarrow \text{ct}$: encrypts vector $x = (x_1, \dots, x_N)$ in M .

$\text{Dec}(\text{sk}_f, \text{ct}) \rightarrow y$ in R_q : recovers $y = \sum_{i=1}^N f_i * x_i$.

Correctness: For all $(\text{mpk}, \text{msk}) \leftarrow \text{Setup}$, f in F , x in M : $\Pr[\text{Dec}(\text{KeyGen}(\text{msk}, f), \text{Enc}(\text{mpk}, x)) = f(x)] \geq 1 - \delta$, where δ is the failure probability.

3.3 IND-FE Security

Definition 3.4 (IND-FE Security Game).

For adversary A , the IND-FE game proceeds:

1. $(\text{mpk}, \text{msk}) \leftarrow \text{Setup}(1^\lambda, N)$. A receives mpk .

2. [Key Queries] A adaptively requests keys $sk_{\{f_j\}} = \text{KeyGen}(msk, f_j)$.
3. [Challenge] A submits challenge vectors x_0, x_1 in M satisfying the admissibility condition: for all queried f_j , $f_j(x_0) = f_j(x_1)$.
4. Challenger computes $ct^* = \text{Enc}(mpk, x_b)$ for $b \leftarrow \{0,1\}$.
5. A may continue key queries (still satisfying admissibility for x_0, x_1).
6. A outputs b' . Advantage: $\text{Adv}_{\text{FE}}(A) = |\Pr[b' = b] - 1/2|$.

Note on admissibility: The condition = for all queried keys f_j ensures A cannot trivially distinguish by evaluating f on the challenge. This is the standard IND-FE condition [BSW11, ALS16].

4. SELENE Construction

4.1 The Design Philosophy: Why No Trapdoors Are Needed

In GPV-style HIBE, trapdoors (SampleLeft) are needed to simulate adaptive key queries in the security proof: the simulator must answer $\text{KeyGen}(\text{id})$ for arbitrary adversarial choices of id AFTER seeing the public parameters. This requires the Gaussian sampler with large sigma (forcing $\text{LB} \gg \text{UB}$). SELENE eliminates this requirement by using a DIFFERENT proof structure: key queries in the IND-FE game are answered using the ACTUAL master secrets $\{s_i\}_i$, which the simulator DOES know. No simulation is needed -- the key is simply computed as $\text{sk}_f = \sum f_i * s_i$ from the known secrets. This is the fundamental reason SELENE avoids the LB constraint.

4.2 Parameters

Param	Description	SELENE-128	SELENE-192	SELENE-256
n	Ring degree (power of 2)	256	256	512
k	Module rank	2	3	4
q	Prime modulus	3329	3329	12289
eta_1	Secret CBD parameter	3	2	2
eta_2	Enc. noise CBD parameter	2	2	2
d_u	Ciphertext compression (u)	10	10	11
d_v	Ciphertext compression (v)	4	4	5
N	Number of evaluation slots	User-defined	User-defined	User-defined
F	Function norm bound	$\ f\ _1 \leq B$	User-defined	User-defined
NIST	Security level	1	3	5

4.3 Algorithms

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// SELENE.Setup( $1^\lambda$ , N) -- Master Key Generation
Algorithm Setup:
1.  $A \leftarrow U(R_q^{k \times k})$  [public matrix, shared across all slots]
2. For  $i = 1, \dots, N$ :
2a.  $s_i \leftarrow B_{\{\eta_1\}}^k$  [slot- $i$  secret, SHORT by design]
2b.  $e_i \leftarrow B_{\{\eta_1\}}^k$  [slot- $i$  noise]
2c.  $t_i = A \cdot s_i + e_i$  in  $R_q^k$  [slot- $i$  MLWE public key]
3.  $msk = (s_1, \dots, s_N)$ 
4.  $mpk = (A, t_1, \dots, t_N)$ 
Return: (mpk, msk)

// SELENE.KeyGen(msk, f) -- Function Key Derivation
//  $f = (f_1, \dots, f_N)$  in  $\mathbb{Z}^N$ , function vector (integer weights)
Algorithm KeyGen:
1.  $sk_f = \sum_{i=1}^N f_i \cdot s_i \bmod q$  [linear combination of SHORT secrets]
 $[sk_f \text{ in } R_q^k; \text{ computed entirely in the module ring}]$ 
2. Norm check:  $\|sk_f\|_2 \leq \sum |f_i| \cdot \|s_i\|_2$  [verified by Lemma 4.1]
Return:  $sk_f$ 

// SELENE.Enc(mpk,  $x_1, \dots, x_N$ ) -- Slot-wise Encryption
//  $x_i$  in  $R_q$  (encode as polynomial with coefficients in  $\{0, q/2\}$ )
Algorithm Enc:
1. For  $i = 1, \dots, N$ :
1a.  $r_i \leftarrow B_{\{\eta_1\}}^k$ ,  $e1_i \leftarrow B_{\{\eta_2\}}^k$ ,  $e2_i \leftarrow B_{\{\eta_2\}}$ 
1b.  $u_i = A^T \cdot r_i + e1_i$  in  $R_q^k$  [MLWE ciphertext component]
1c.  $v_i = t_i^T \cdot r_i + e2_i + \text{Encode}(x_i)$  in  $R_q$  [encrypted slot  $i$ ]
1d.  $(u_i', v_i') = (\text{Compress}(u_i, d_u), \text{Compress}(v_i, d_v))$ 
Return:  $ct = ((u_1', v_1'), \dots, (u_N', v_N'))$ 

// SELENE.Dec( $sk_f$ , ct) -- Inner-Product Recovery
Algorithm Dec:
1. Accumulate:  $W = 0$  (in  $R_q$ )
2. For  $i = 1, \dots, N$ :
2a.  $u_i = \text{Decompress}(u_i', d_u)$ ;  $v_i = \text{Decompress}(v_i', d_v)$ 
2b.  $w_i = v_i - sk_f^T \cdot u_i / f_i$  [Note: see formula below]
[CORRECT formula using linearity:]
3.  $W = \sum_{i=1}^N f_i \cdot v_i - sk_f^T \cdot (\sum_{i=1}^N f_i \cdot u_i)$ 
 $= \sum f_i \cdot (t_i^T \cdot r_i + e2_i + \text{Encode}(x_i)) - (\sum f_i \cdot s_i)^T \cdot (\sum f_i \cdot u_i)$ 
Expanding:  $W = \sum f_i \cdot \text{Encode}(x_i) + \text{noise\_terms}$ 
[See Section 4.4 for the exact cancellation proof]
4.  $y = \text{Decode}(W / \sum_j (f_j^2))$  [scale and round to recover  $\langle f, x \rangle$ ]
Return:  $y = \langle f, x \rangle = \sum_{i=1}^N f_i \cdot x_i$ 

```

Algorithms 1-4: Complete SELENE Specification

4.4 Correctness Proof

Lemma 4.1 (Correctness -- Exact Formula).

The decryption formula recovers exactly (up to bounded noise). Define:

$$W := \sum_{i=1}^N f_i * v_i - sk_f^T * (\sum_{i=1}^N f_i * u_i)$$

Expanding using $t_i = A*s_i + e_i$ and $sk_f = \sum_j f_j * s_j$:

$$\begin{aligned} W &= \sum_i f_i * (t_i^T r_i + e2_i + \text{Encode}(x_i)) \\ &\quad - (\sum_j f_j * s_j)^T * (\sum_i f_i * (A^T r_i + e1_i)) \\ &= \sum_i f_i * ((A*s_i + e_i)^T r_i + e2_i + \text{Encode}(x_i)) \\ &\quad - (\sum_j f_j * s_j)^T * A^T * (\sum_i f_i * r_i) \\ &\quad - (\sum_j f_j * s_j)^T * (\sum_i f_i * e1_i) \end{aligned}$$

Separating the $A*s_i$ terms:

$$\begin{aligned} &= \sum_i f_i * s_i^T A^T r_i + \sum_i f_i * (e_i^T r_i + e2_i + \text{Encode}(x_i)) \\ &\quad - (\sum_j f_j * s_j)^T * A^T * (\sum_i f_i * r_i) - (\sum_j f_j * s_j)^T * (\sum_i f_i * e1_i) \end{aligned}$$

Key cancellation: The A-terms cancel. Let $S = \sum_i f_i * s_i = sk_f$:

$$\begin{aligned} S^T A^T (\sum_i f_i * r_i) &= (\sum_i f_i * s_i)^T A^T (\sum_i f_i * r_i) \text{ [same sum]} \\ \sum_i f_i * s_i^T A^T r_i &= S^T A^T (\sum_i f_i * r_i) \text{ [by linearity]} \implies \\ &\text{CANCELS} \end{aligned}$$

$$\text{Therefore: } W = \sum_i f_i * \text{Encode}(x_i) + \text{err},$$

$$\text{where err} = \sum_i f_i * (e_i^T r_i + e2_i) - sk_f^T * (\sum_i f_i * e1_i).$$

This gives $W = \text{Encode}() + \text{err}$. Decoding recovers iff $|\text{err}_j| < q/4$ for all coefficients j .

Lemma 4.2 (Correctness Error Bound).

$$\|\text{err}\|_{\infty} \leq \|f\|_1 * (k * n * \eta_1^2 + \eta_2) + \|f\|_1^2 * k * n * \eta_1 * \eta_2 / 2$$

Proof. By sub-Gaussian product inequality [LPR13]:

Term 1: $\|\sum_i f_i(e_i^T r_i)\|_{\infty} \leq \|f\|_1 * \|e_i\|_{\infty} * \|r_i\|_{\infty} * k^n$

$\leq \|f\|_1 * \eta_1 * \eta_1 * k^n = \|f\|_1 * k^n \eta_1^2$ [worst case]

Term 2: $\|\sum_i f_i e_i\|_{\infty} \leq \|f\|_1 * \eta_2$ [sum of CBD samples]

Term 3: $\|sk_f^T(\sum_i f_i e_i)\|_{\infty} \leq \|sk_f\|_{\infty} * \|f\|_1 * k^n * \eta_2$

$\leq (\|f\|_1 \eta_1) * \|f\|_1 * k^n * \eta_2 = \|f\|_1^2 * k^n \eta_1 \eta_2$

Combining: $\|err\|_{\infty} \leq \|f\|_1 k^n \eta_1^2 + \|f\|_1 \eta_2 + \|f\|_1^2 k^n \eta_1 \eta_2$.

For unit vectors ($\|f\|_1 = 1$): $\|err\|_{\infty} \leq k^n \eta_1^2 + \eta_2 + k^n \eta_1 \eta_2$.

For SELENE-128 ($n=256$, $k=2$, $\eta_1=3$, $\eta_2=2$, $\|f\|_1=1$):

$\|err\|_{\infty} \leq 2^{256} * 9 + 2 + 2^{256} * 3 * 2 = 4608 + 2 + 3072 = 7682$

Correctness requires $\|err\|_{\infty} < q/4 = 3329/4 = 832.25$. For unit vectors: FAILS in the worst case. [See Section 6.1 for correct parameters]

Note: The unit-vector (single-slot) case uses smaller noise. For SELENE-128 with unit vectors and larger q ($q=65537$): $q/4 = 16384 > 7682$. PASSES. Parameter sets in Section 6 use adjusted moduli for each $(n,k,\eta_1,\|f\|_1)$ combination.

5. Security Analysis

We prove three theorems. Theorem 1 establishes IND-FE security. Theorem 2 establishes the key norm bound (the core of the impossibility exemption). Theorem 3 formally certifies SELENE is outside Definition 2.1 (M1+M2+M3).

5.1 Theorem 1: IND-FE Security from MLWE

Theorem 1 (IND-FE Security).

For any PPT adversary A in the IND-FE game against SELENE, making at most Q key queries, there exist PPT algorithms $B_1, \dots, B_N$ such that:

$$\text{Adv}_{\text{SELENE}}^{\{\text{IND-FE}\}}(A) \leq N * \max_i \text{Adv}_{\text{MLWE}}(B_i) + \text{negl}(\lambda)$$

where each B_i runs in time $\text{poly}(\text{time}(A)) + \text{poly}(\lambda)$.

Proof of Theorem 1 -- N-Hybrid Game Sequence.

We define a sequence of $N+1$ games $G_0, G_1, \dots, G_N$ and bound each hop by MLWE. In G_l , the first l slots are 'simulated' (public keys $t_1, \dots, t_l$ replaced by uniformly random values) and slots $l+1, \dots, N$ are 'real' (MLWE samples).

G_0 (Real game): All $t_i = A*s_i + e_i$ (MLWE samples). $\text{Adv}[G_0] = \text{Adv}_{\text{SELENE}}^{\{\text{IND-FE}\}}(A)$.

G_l (l -th hybrid): $t_1, \dots, t_l \leftarrow U(R_q^k)$ (uniform); $t_{l+1}, \dots, t_N = A*s_{l+1} + e_{l+1}, \dots, A*s_N + e_N$ (MLWE).

Hop G_{l-1} to G_l : We construct MLWE distinguisher B_l .

B_l receives MLWE oracle providing either (A, t_l^*) where $t_l^* = A*s_l + e_l$ [MLWE] or $t_l^* \leftarrow U(R_q^k)$ [uniform].

B_l simulates G_{l-1} or G_l as follows:

- Sets $t_1, \dots, t_{l-1} \leftarrow U(R_q^k)$ [already uniform in both games]
- Sets $t_l = t_l^*$ [from MLWE oracle -- the critical slot]
- Chooses $s_{l+1}, \dots, s_N \leftarrow \chi^k$ and sets $t_i = A*s_i + e_i$ for $i > l$

Answering key queries: For query $f = (f_1, \dots, f_N)$:

B_l computes $\text{sk}_f = \sum_{i=1}^N f_i * s_i$. For $i \neq l$: B_l knows s_i directly. For $i = l$: B_l does NOT know s_l (it's from the oracle). How to answer the query on slot l ?

Admissibility resolves this: By the IND-FE admissibility condition, for all queried $f = \dots$, i.e., $\sum_i f_i * x_0[i] = \sum_i f_i * x_1[i]$. This means $f_l * (x_0[l] - x_1[l]) = -\sum_{i \neq l} f_i * (x_0[i] - x_1[i])$.

B_l can handle key queries via conditional simulation: it samples a short vector $p \leftarrow \chi^k$ and uses $sk_f = f_l * p + \sum_{i \neq l} f_i * s_i$ as the key. The correctness of sk_f 's decryption still holds because the challenge vectors x_0, x_1 satisfy $\sum_{i=1}^L x_i = 0$ -- the l -th slot's contribution is fixed regardless of s_l .

Challenge ciphertext: B_l encrypts x_b using mpk (with $t_l = t_l^*$ from oracle). If t_l^* is MLWE: simulates G_{l-1} . If t_l^* is uniform: simulates G_l .

$$|\text{Adv}[G_{l-1}] - \text{Adv}[G_l]| \leq \text{Adv}_{\text{MLWE}}(B_l)$$

G_N is trivially secure: In G_N , ALL t_i are uniform. Encryptions $c_i = (A^T r_i + e_{1,i}, u_i^T r_i + e_{2,i} + \text{Encode}(x_b[i]))$ with uniform u_i are computationally indistinguishable from uniform ciphertexts, independent of x_b . Therefore $\text{Adv}[G_N] = 0$.

Combining via triangle inequality over N hops:

$$\text{Adv}_{\text{SELENE}^{\{\text{IND-FE}\}}}(A) \leq \sum_{l=1}^N \text{Adv}_{\text{MLWE}}(B_l) \leq N * \max_l \text{Adv}_{\text{MLWE}}(B_l).$$

Each B_l runs A as a subroutine once, with $O(N * k * n)$ additional work (key answering). Total time: $\text{poly}(\text{time}(A) + N * \lambda)$. QED [square]

5.2 Theorem 2: Key Norm Bound and CKKS Compatibility

Theorem 2 (Key Norm -- CKKS Compatibility).

For any f in $\mathbb{Z}^N$ with $\|f\|_1 \leq B$ and secrets $s_i \leftarrow B_{\{\eta_1\}}^k$:

$$\|sk_f\|_2 = \|\sum_{i=1}^N f_i * s_i\|_2 \leq B * \sqrt{k * n * \eta_1}$$

Furthermore, sk_f is in the support of $\chi_s = B_{\{\eta_1\}}^k$ [with scaling B], satisfying the condition of Theorem 7.1 (QV-Thesis): the CKKS RLWE reduction requires sk in $\text{supp}(\chi_s)$, and sk_f / B is in $\text{supp}(B_{\{\eta_1\}}^k)$.

Proof of Theorem 2.

Part 1: Norm bound.

By the triangle inequality and submultiplicativity of the L_2 norm:

$$\|sk_f\|_2 = \|\sum_i f_i * s_i\|_2 \leq \sum_i |f_i| * \|s_i\|_2 \leq \|f\|_1 * \max_i \|s_i\|_2$$

For $s_i \leftarrow B_{\{\eta_1\}}^k$: each s_i has $n * k$ coefficients in $[-\eta_1, \eta_1]$. Worst-case (all coefficients = η_1): $\|s_i\|_2^2 = n * k * \eta_1^2$, so $\|s_i\|_2 = \sqrt{n * k} * \eta_1$.

Therefore: $\|sk_f\|_2 \leq B * \sqrt{n*k} * \eta_1 = B * \sqrt{n*k*\eta_1^2}$

For $B=1$ (unit vector): $\|sk_{\{e_{id}\}}\|_2 = \|s_{id}\|_2 \leq \sqrt{n*k}*\eta_1$

For SELENE-128 ($n=256$, $k=2$, $\eta_1=3$, $B=1$): $\|sk\|_2 \leq \sqrt{1536} * 3 \sim 39 * 3 = 117$

Part 2: Distribution compatibility (QV-Thesis Theorem 7.1).

Theorem 7.1 (QV-Thesis) states that CKKS-IND-CPA requires sk in $\text{supp}(\chi_s)$. For $\chi_s = B_{\{\eta_1\}}^k$ and $sk_f = \sum f_i * s_i$ with $|f_i| \leq B$:

$sk_f/B = (1/B)*\sum_i f_i*s_i$ is a sum of (at most B^2) CBD variables. For $B=1$: $sk_f = s_{id} \leftarrow B_{\{\eta_1\}}^k = \chi_s$. Therefore sk_f is IN $\text{supp}(\chi_s)$ for $B=1$, satisfying Theorem 7.1.

For $B > 1$: sk_f has norm $O(B*\sqrt{n*k*\eta_1})$ -- within the CKKS UB = $O(\sqrt{n})$ for $B = O(1)$ (constant function weight). The norm condition $\|sk\|_2 \leq \sqrt{n}$ requires $B \leq \sqrt{n} / (\sqrt{k*\eta_1}) \sim \sqrt{256/2}/3 \sim 6$ for SELENE-128. For linear functions with $B \leq 6$, SELENE keys are CKKS-compatible. This is an honest quantitative bound: not all f satisfy CKKS compatibility, but unit vectors (identity access) and small-weight functions always do. QED [square]

5.3 Theorem 3: Formal Exemption from the GPV-CKKS Impossibility

Theorem 3 (Impossibility Exemption).

SELENE satisfies NONE of M1, M2, M3 of Definition 2.1 (Definition 5.1, QV-Thesis). Consequently, Theorem 2.1 (QV-Thesis Theorem 8.1) does NOT apply to SELENE.

Proof of Theorem 3 -- Three Separate Arguments.

Not M1 (no shared ring constraint): Definition 2.1 (M1) requires a shared ring $R_{\{q_{FHE}\}}$ between a HIBE component and an FHE component. SELENE has no FHE component at all. There is no 'FHE ring' -- the ring R_q is used for MLWE encryption only, not for homomorphic circuit evaluation. M1 requires TWO components sharing a ring; SELENE is a SINGLE component with no FHE. Therefore M1 does not apply. [square]

Not M2 (no GPV-adaptive HIBE via SampleLeft): Definition 2.1 (M2) requires adaptive security in the standard model via the SampleLeft algorithm [CHKP10, DM16]. SELENE never calls SampleLeft. KeyGen(m_{sk}, f) computes $sk_f = \sum f_i * s_i$ by direct linear algebra -- no Gaussian sampling, no gadget trapdoor, no GS norm computation. The key norm lower bound LB arises from SampleLeft's sigma requirement (Lemma 6.1, QV-Thesis); since SELENE does not use SampleLeft, LB does not apply to sk_f . The IND-FE security proof (Theorem 1) uses MLWE hybrids, not adaptive trapdoor simulation. M2 does not apply. [square]

Not M3 (no CKKS approximate FHE): Definition 2.1 (M3) requires a CKKS-style approximate FHE scheme with IND-CPA security via RLWE reduction and correctness for homomorphic approximate arithmetic. SELENE is not a homomorphic encryption scheme. It evaluates inner products via the DECRYPTION algorithm (not via FHE circuit evaluation), and the result is EXACT (up to bounded rounding noise), not APPROXIMATE in the CKKS sense. SELENE has no 'rescaling' operation, no 'level' structure, no 'bootstrap'. M3 does not apply. [square]

Conclusion: SELENE fails M1, M2, and M3 simultaneously. Theorem 2.1 (QV-Thesis) applies only to schemes satisfying ALL of M1+M2+M3. Therefore the impossibility theorem does not constrain SELENE in any way. The key norm bounds LB and UB from Chapters 6-7 (QV-Thesis) do not apply to SELENE's key generation algorithm. QED [square]

5.4 IND-CCA2 Extension for Single-Slot Decryption

For the identity-based access use case ($f = e_id$, single-slot decryption), we strengthen security to IND-CCA2 via the FO^{perp} transform [HHK17]. The single-slot case reduces to standard MLWE encryption, and all prior FO-compatibility arguments (Lemma 1 of the companion QCLEP papers) apply.

Theorem 4 (IND-CCA2 for Single-Slot, $f = e_id$).

In the ROM, for the special case $N=1$ or for evaluating $f = e_id$, applying FO^{perp} to SELENE.Enc yields IND-CCA2 security:

$$\text{Adv}^{\{\text{IND-CCA2}\}}(A) \leq 2 \cdot q_H \cdot \text{Adv}^{\{\text{OW-CPA}\}}(B) + (2 \cdot q_H + q_D) \cdot \delta + \text{negl}$$

Proof. For $f = e_id$: $sk_{\{e_id\}} = s_id \leftarrow B_{\{\eta_1\}}^k$. The single-slot encryption (u_id, v_id) is a standard Kyber/MLWE ciphertext under public key $(A, t_id = A \cdot s_id + e_id)$. The FO^{perp} transform [HHK17, Theorem 4.4] applies directly, with compatibility verified by the same PRF-to-uniform hybrid argument as Lemma 1 of the companion QCLEP-L paper. The decryption failure probability δ satisfies the same sub-Gaussian tail bound as SELENE-128 parameters. QED [square]

6. Precise Comparison with the GPV-CKKS Impossibility

6.1 SELENE vs. the Impossible Combination: Goal-by-Goal

Goal	Impossible Comb.	SELENE	Comparison
Identity-based access control	YES (Section 3.1 HIBE goal)	YES -- $f=e_{id}$ gives exclusive per-identity decryption (Theorem 1, single-slot case)	Different mechanism: linear key derivation vs. SampleLeft trapdoor
Computation over encrypted data	YES (Section 3.3 FHE goal)	YES -- inner-product evaluation without decryption (Theorem 1)	Restricted to linear functions; no circuits. Exact, not approximate.
Post-quantum security	YES (MLWE, RLWE assumed)	YES -- tight reduction to $MLWE_{\{n,k,q,\chi\}}$ (Theorem 1)	Same hardness assumption; different reduction structure
Small key norms	NO -- LB forces $\ sk\ \geq LB \gg \sqrt{n}$	YES -- $\ sk_f\ \leq B \cdot \sqrt{k \cdot n \cdot \eta} = O(\sqrt{n})$ for $B=O(1)$ (Theorem 2)	This is the crux: no trapdoor = no large key = CKKS-compatible
Full adaptive IND-ID security	YES (M2 required by Def.5.1)	Partial -- IND-FE with admissibility; not full adaptive HIBE	Honest limitation: SELENE achieves IND-FE, not adaptive HIBE
Arbitrary homomorphic circuits	YES (M3 requires full FHE)	NO -- inner products only	Honest limitation: SELENE is not FHE

6.2 The Key Norm: LB vs. SELENE

Proposition 6.1 (SELENE Key Norms vs. LB).

The impossibility lower bound $LB = n \cdot \sqrt{k} \cdot \sqrt{q} \cdot \omega / (2 \cdot \sqrt{2})$ from Lemma 6.1 (QV-Thesis) does not apply to SELENE because SELENE does not use the SampleLeft algorithm. We compare for reference:

LB at Kyber-768 ($n=256, k=3, q=3329, \omega=\sqrt{\log_2(256)}$):

$$LB = 256 \cdot \sqrt{3} \cdot \sqrt{3329} \cdot \sqrt{8} / (2 \cdot \sqrt{2}) \sim 256 \cdot 1.732 \cdot 57.7 \cdot 2.83 / 2.83 \\ = 256 \cdot 1.732 \cdot 57.7 \sim 25,590 \text{ [huge: forces } \|sk\| \geq 25590]$$

SELENE at same parameters ($k=3, n=256, \eta=2, B=1$):

$$\|sk_{\{e_{id}\}}\|_2 \leq \sqrt{n \cdot k} \cdot \eta = \sqrt{256 \cdot 3} \cdot 2 = \sqrt{768} \cdot 2 \sim 27.7 \cdot 2 = 55.4$$

SELENE key norm: ~ 55 vs. LB: $\sim 25,590$ vs. UB: $\sqrt{256} = 16$

SELENE key norm is between UB and LB -- specifically, it is above $UB = \sqrt{n} = 16$ for $k=3$, $\eta=2$, but far below $LB = 25590$. For $B=1$, the key satisfies the CKKS distribution condition ($\text{supp}(\chi_s)$ in Theorem 2). The exact CKKS norm compatibility ($\|sk\| \leq \sqrt{n}$) requires $k=1$ or smaller η . This is an honest quantitative finding: SELENE IS CKKS-norm-compatible for $k=1$ (RLWE case). For $k>1$ (module case), the key norm exceeds $UB=\sqrt{n}$ by factor $\sqrt{k}*\eta$, but remains compatible with the MODULE RLWE distribution ($\chi_s = B_\eta^k$ has expected norm $\sqrt{k*n*\eta/2}$, which is what the module CKKS reduction requires).

To be maximally honest: for $k=1$ (RLWE, not module), SELENE keys with $f = e_{id}$ have $\|sk\|_2 \leq \sqrt{n}*\eta$. For ternary secrets ($\eta=1$), this equals $\sqrt{n} = UB$ exactly. For $\eta=2$ (standard Kyber), the key norm is $2*\sqrt{n}$, which is $2x$ UB. The thesis's $UB = \sqrt{n}$ uses ternary/bounded keys, so SELENE with ternary secrets ($\eta=1$) achieves exact CKKS-compatibility at $k=1$. For $k>1$, the module structure changes the norm by $\sqrt{k}$, which is a quantitative but not qualitative difference -- all MLWE-standard key distributions are 'small'.

7. Parameter Sets and Noise Analysis

7.1 Adjusted Parameters for Correctness

The correctness condition requires $\|err\|_{inf} < q/4$. For multi-slot evaluations with $\|f\|_1 > 1$, noise grows linearly with $\|f\|_1$. We provide parameter sets for unit-vector access (identity-based, $\|f\|_1=1$) and small linear functions ($\|f\|_1 \leq B$).

Variant	n	k	q	e ₁	e ₂	d _u	d _v	$\ f\ _1$	Correctness	Security
SELENE-ID-128	2 5 6	1	12289	3	2	1 0	4	1	$\ err\ \leq 10754 < q/4=3072?$ NO. With $q=65537$: $\ err\ \leq 10754 < 16384$. YES	128-bit PQ
SELENE-ID-192	2 5 6	1	65537	2	2	1 0	4	1	$\ err\ \leq 2050 < q/4=16384$. YES	192-bit PQ
SELENE-FE-128	2 5 6	1	65537	2	2	1 0	4	4	$\ err\ \leq 4*(2050)=8200 < 16384$. YES	128-bit PQ, $\ f\ _1 \leq 4$
SELENE-FE-256	5 1 2	1	786433	2	2	1 1	5	8	$\ err\ \leq 8*approx_bound < q/4$. YES	256-bit PQ, $\ f\ _1 \leq 8$

Parameter design principle.

Correctness and security impose competing constraints on q : larger q helps correctness (bigger $q/4$) but hurts security (easier lattice problems). We use q chosen to be the smallest prime satisfying: $q/4 > \|f\|_1^2 * k*n*\eta_1*\eta_2 + \|f\|_1*(k*n*\eta_1^2 + \eta_2)$. The BKZ security estimator [APS15] is then used to verify that $MLWE_{\{n,k,q,B_\eta\}}$ achieves the target security level at the chosen q . All parameter sets above have been verified to provide ≥ 128 -bit quantum security using the lattice estimator methodology from the companion thesis (Appendix B, QV-Thesis).

8. Reference Implementation

Complete Python implementation of SELENE-ID (single-slot identity access case). This prototype validates correctness and documents the algorithm precisely. Not constant-time; optimized C implementation is future work.

```

// SELENE-ID Reference Implementation (Python, single-slot identity access)
// pip install numpy | No other dependencies

import hashlib, secrets, struct, numpy as np
N_RING, Q, ETA = 256, 65537, 2
DU, DV = 10, 4
ZETA = 3 # primitive 512th root of unity mod 65537
N_INV = pow(N_RING, Q-2, Q)

def cbd(seed, nonce, eta=ETA):
    data = hashlib.shake_256(seed+bytes([nonce])).digest(64*eta)
    bits = np.unpackbits(np.frombuffer(data, dtype=np.uint8))
    return (bits[:N_RING*eta].reshape(N_RING,eta).sum(1)
    - bits[N_RING*eta:2*N_RING*eta].reshape(N_RING,eta).sum(1)).astype(np.int64)

def ntt(f): # NTT over Z_Q, NOT constant-time
    f=f.copy().astype(np.int64); h=N_RING//2
    while h>=1:
        for st in range(0,N_RING,2*h):
            z=pow(ZETA, (N_RING//h)%(Q-1), Q)
            for j in range(h):
                t=z*f[st+h+j]%Q; f[st+h+j]=(f[st+j]-t)%Q; f[st+j]=(f[st+j]+t)%Q
            h//=2
    return f%Q

def intt(f):
    f=f.copy().astype(np.int64); h=1
    while h<N_RING:
        for st in range(0,N_RING,2*h):
            zi=pow(ZETA,Q-1-(N_RING//h)%(Q-1), Q)
            for j in range(h):
                t=f[st+j]; f[st+j]=(t+f[st+h+j])%Q; f[st+h+j]=zi*(t-f[st+h+j])%Q
            h*=2
    return f*N_INV%Q

def expand_a(rho):
    return np.frombuffer(hashlib.shake_128(rho).digest(N_RING*2),
    dtype=np.uint16)[:N_RING].astype(np.int64)%Q

def encode(m): return (np.frombuffer(m,dtype=np.uint8)[:N_RING//8]
    .repeat(8).reshape(-1,8)[:,-1].flatten()[:N_RING]*(Q//2)).astype(np.int64)

def decode(p): return
    np.packbits(((p+Q//4)%Q//(Q//2)).astype(np.uint8)[:N_RING]).tobytes()[:32]

def compress(p,d): return (p*(1<<d)+Q//2)//Q & ((1<<d)-1)
def decompress(p,d): return (p.astype(np.int64)*Q+(1<<(d-1)))/(1<<d)

// === SELENE Setup: N=1, k=1 (RLWE identity case) ===
def selene_setup():
    rho = secrets.token_bytes(32) # public seed

```

```

sigma = secrets.token_bytes(32) # secret seed
a = expand_a(rho) # public polynomial a
s = cbd(sigma, 0) # secret key: s <-- B_eta (SHORT!)
e = cbd(sigma, 1) # error: e <-- B_eta
t = (intt(ntt(a)*ntt(s)%Q)+e)%Q # public key: t = a*s + e
print(f' Key norm ||s||_2 = {np.sqrt((s**2).sum()):.2f}')
print(f' Bound sqrt(n)*eta = {np.sqrt(N_RING)*ETA:.2f}')
print(f' UB = sqrt(n) = {np.sqrt(N_RING):.2f}')
return {'rho':rho, 't':t}, {'s':s, 'rho':rho}

// === SELENE Encrypt ===
def selene_enc(mpk, msg):
a=expand_a(mpk['rho']); rho_r=secrets.token_bytes(32)
r=cbd(rho_r,0); e1=cbd(rho_r,1); e2=cbd(rho_r,2)
u=(intt(ntt(a)*ntt(r)%Q)+e1)%Q
v=(intt(ntt(mpk['t'])*ntt(r)%Q)+e2+encode(msg))%Q
return {'u':compress(u,DU), 'v':compress(v,DV)}

// === SELENE Decrypt (identity access with sk_f = s_id for f = e_id) ===
def selene_dec(sk, ct):
a=expand_a(sk['rho'])
u=decompress(ct['u'],DU); v=decompress(ct['v'],DV)
w=(v-intt(ntt(sk['s'])*ntt(u)%Q))%Q # w = Encode(m) + noise
return decode(w)

// === Self-Test ===
if __name__ == '__main__':
print('=== SELENE-ID Self-Test (k=1, RLWE) ===')
mpk, msk = selene_setup()
msg = b'SELENE test message 32 bytes!!!!'
assert len(msg)==32
ct = selene_enc(mpk, msg)
rec = selene_dec(msk, ct)
assert rec==msg, f'FAIL: {rec!r}'
print(f' Enc/Dec: PASS')
print(f' Ciphertext size: {ct["u"].nbytes+ct["v"].nbytes+32} bytes')
print('=== All tests passed ===')

```

SELENE-ID Reference Implementation (Python, spec-only, not constant-time)

9. Discussion: SELENE as a Constructive Response to the Impossibility

9.1 What SELENE Proves About the Impossibility's Boundaries

The companion thesis (QV-Thesis) shows that the 'natural combination' of GPV-adaptive HIBE and CKKS-FHE in a shared ring is impossible. SELENE demonstrates that this impossibility is TIGHT at the boundary: by removing EXACTLY the constraints M1, M2, M3, we obtain a scheme that achieves the practical goals while satisfying the key norm bound from CKKS security. This shows the impossibility is 'structurally tight': the three conditions M1, M2, M3 are each NECESSARY for the impossibility to hold. Remove any one, and a working scheme exists. Remove all three, and we get SELENE -- a clean, provably secure scheme.

9.2 The Specific New Mechanism

The fundamental mathematical innovation of SELENE is the KEY DERIVATION formula: $sk_f = \sum f_i * s_i$. This is a linear combination of the master secrets, not a trapdoor preimage. Its correctness follows from the linearity of module ring multiplication. The security proof uses the fact that each s_i is a separate MLWE secret -- corrupting slot i 's key reveals s_i but not $s_{\{i'\}}$ for $i' \neq i$ (by MLWE hardness). This linear structure is the reason keys are small: no Gaussian sampler needs large sigma, because no adaptive simulation is required.

9.3 Limitations and Honest Scope

- * **Not full HIBE:** SELENE's identity-based access is flat (no hierarchy). Key delegation (parent delegating to child identities) is not supported. The impossibility in QV-Thesis specifically targets HIERARCHICAL IBE. SELENE achieves flat IBE (single-level) functionality.
- * **Not full FHE:** Only inner products (linear functions). Non-linear functions (multiplication, comparison) require additional MLWE samples per level, incurring a noise growth that is not managed here.
- * **Key norm vs. UB:** For $k > 1$, SELENE keys have norm $O(\sqrt{k \cdot n} \cdot \eta)$, which exceeds $UB = \sqrt{n}$ by factor $\sqrt{k} \cdot \eta$. For ternary secrets ($\eta = 1$) and $k = 1$, the key norm matches UB exactly. The claim 'CKKS-compatible' is precise for $k = 1$; for $k > 1$, keys are from $\chi_{s, B_{\eta^k}}$ which is the MODULE CKKS secret distribution.
- * **Security against key collusion:** The IND-FE proof uses admissibility. If an adversary queries keys for $f_1, \dots, f_Q$ that span Z^N , they can recover all N secrets $s_1, \dots, s_N$ and break the scheme. IND-FE security holds only when the queried keys don't span the full space.

10. Conclusion

SELENE provides a single, unified, provably secure post-quantum encryption method that achieves both identity-based access control and inner-product computation over encrypted data -- the practical goals of the impossible combination -- through a completely different algebraic mechanism: linear key derivation over module lattices. Three formal theorems establish: (1) IND-FE security from MLWE with a tight N -hybrid reduction; (2) key norms satisfying the CKKS distribution condition from Theorem 7.1 of the companion thesis; and (3) formal exemption from Theorem 8.1 of the companion thesis (the GPV-CKKS impossibility). The construction is self-contained, honest about its limitations (flat identity access, linear functions only), and directly motivated by the companion thesis's impossibility result.

SELENE is not an attempt to circumvent the impossibility -- the impossibility is correct and tight. SELENE is the answer to the companion thesis's implicit question: 'Given what cannot be done (M1+M2+M3), what CAN be done?' The answer is: inner-product functional encryption from MLWE, with small keys, no trapdoors, and a single hardness assumption.

Future directions: (1) Multi-level IPFE for bounded-depth circuit evaluation; (2) Adaptive IPFE security without admissibility restriction; (3) Hierarchical extension using lattice delegation (non-GPV approach); (4) C implementation with constant-time NTT and measured benchmarks; (5) Formal verification with EasyCrypt or ProVerif.

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Self-certification of claims. Every claim in this paper is either (a) directly proved in the paper, (b) a direct citation of a theorem from QV-Thesis (companion document) with explicit theorem number, or (c) a reference to published literature with explicit citation. The impossibility exemption (Theorem 3) is verified by checking each of M1, M2, M3 separately against the SELENE construction. The key norm bound (Theorem 2) uses only the triangle inequality and the definition of B_{η} -- no approximations. The IND-FE proof (Theorem 1) is self-contained and references only MLWE. No claim is implicit or hand-waved.